

A "Second Life" for Rescued Bread: A Life Cycle Assessment

Environmental Impact Comparison:
Rescued Bread Flour vs. Conventional Wheat Flour

Our Mission: Why Rescue Bread?

Food waste is a critical global issue. In New Zealand, bread is the second most wasted food item in households (accounting for 29%).

Rescued NZ offers an innovative solution by upcycling unsold surplus bread from supermarkets into high-quality "Rescued Bread Flour".

This study aims to scientifically quantify the environmental footprint of **Rescued Bread Flour** compared to **Conventional Wheat Flour**.

Key Findings: A Clear Win for Rescued Bread Flour!



Impact Category	Unit	Rescued Flour	Wheat Flour	Change
Carbon Footprint	kg CO ₂ eq	377	1786	↓ 79.0%
Land Use	m ² a crop eq	16	6000	↓ 99.9%
Fossil Fuel Use	kg oil eq	99	346	↓ 31.0%
Freshwater Toxicity	kg 1,4-DCB	10	25	↓ 60.0%
Air Pollution (PM2.5)	kg PM _{2.5} eq	0.25	3.67	↓ 93.0%

The results clearly show that Rescued Bread Flour significantly outperforms Conventional Wheat Flour in all key environmental impact categories.

The donut charts at the top highlight key improvements: a 99.9% reduction in Land Use, a 79% reduction in Carbon Footprint and 31.0% reduction in Fossil Fuel Use.

The two tables below provide the detailed data:

- Midpoint Impacts: Confirm superiority across all five categories, including Air Pollution (-93%) and Freshwater Toxicity (-60%).

- Endpoint Impacts: Quantify the final benefits, showing a 98% reduction in damage to the natural environment and an 85% reduction in damage to human health.

Future Recommendations

- Energy Transition: Electrify the drying equipment and switch to renewable energy supplier.

- Packaging Innovation: Explore a reuse system for paper bags to reduce forestry resource consumption.

Our Team: 6M Sustainability Consultant



Methodology: The "Life" of Two Flours

We compared the "Cradle-to-Gate" environmental impacts of two products fulfilling the same function: **"Supplying 1 tonne of baking flour for baking purposes, in the Auckland region (NZ)".**

Product A: Rescued Bread Flour

- Raw Material:** Surplus bread from Auckland supermarkets.
- Process:** Local collection, de-bagging, drying, milling, sieving, packaging.

Product B: Conventional Wheat Flour

- Raw Material:** Wheat grown in and imported from Australia.
- Process:** Farming, harvesting, long-distance transport, milling in NZ, packaging.

Conclusion: The Huge Environmental Benefit of Upcycling

By upcycling surplus bread that was destined for waste, Rescued NZ not only reduces food waste but also massively lowers the environmental footprint of a daily staple.

CHOOSING RESCUED BREAD FLOUR IS CHOOSING A KINDER FUTURE FOR OUR PLANET!

Continuous Improvement: They Can Do Even Better

The LCA also helped identify the main environmental hotspots in the Rescued Bread Flour process:

- Drying Process: The main source of Carbon Footprint (85%) and Fossil Fuel Use (82%).

- Packaging: The main source of Land Use (86%).